CONFIDENTIAL

UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION SEMESTER 2, 2023/2024

SUBJECT CODE	**:**	**SECR2043**
SUBJECT NAME	**:**	**OPERATING SYSTEMS**
SECTION	**:**	
TIME	**:**	**3 HOURS**
DATE/DAY	**:**	
VENUES	**:**	

INSTRUCTIONS:

Answer all questions from PART A, and PART B in the answer booklet. Show all your works.

(Please Write Your Lecturer Name And Section In Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section	
Lecturer Name	

This question paper consists of **thirteen (13)** printed pages excluding this page.

PART A: OBJECTIVES

Answer ALL questions in the answer booklet.

1. Deadlock can arise under the following conditions:

i. Mutual Exclusion
ii. Hold and Wait
iii. Preemption
iv. Circular Wait

A) i, ii, iii

B) ii, iii, iv

C) i, iii, iv

D) i, ii, iv

2. Select the following which are part of resource utilization for a process?

i. Delete
ii. Use
iii. Release
iv. Wait

A) i, ii, iii

B) i, ii, iv

C) ii, iii

D) iii, iv

3. Select the statement which is TRUE.

A) Resources can either be physical or logical.

B) Resources can only be physical.

C) Resources can only be logical.

D) Resources can be partitioned into several types using Banker's algorithm.

4. Choose the statement that best describes safe state in a system.

A) A safe state is when there exists a sequence <$PP_0, PP_1, \ldots, PP_{nn}$> of all processes in the system such that for each PP_{ii}, the resources that PP_{ii} can still request can be satisfied by currently available resources and resources held by all PP_{jj} where $jj < ii$.

B) A safe state is when there exists a sequence <$PP_0, PP_1, \ldots, PP_{nn}$> of all processes in the system such that all process requests can be satisfied regardless of whether or not the resource is available.

C) A safe state is when there exists a sequence <$PP_0, PP_1, \ldots, PP_{nn}$> of all processes in the system such that for each PP_{ii}, resources are allocated equally.

D) A safe state is when there exists a sequence <$PP_0, PP_1, \ldots, PP_{nn}$> of all processes in the system such that for each PP_{ii}, the resources are allocated based on the priority of the processes.

5. Which of the following are TRUE statements of fixed partitions?

i. It is a type of contiguous memory allocation strategy.
ii. Free partitions are combined.
iii. It can lead to external fragmentation.
iv. The number of processes in memory is determined by the number of partitions.

A) i and ii
B) ii and iii
C) iii and iv
D) i and iv

6. _____ is the concept in which a process is copied into the main memory from the secondary memory according to the requirement.

A) Demand segmentation
B) Demand paging
C) Segmentation
D) Thrashing

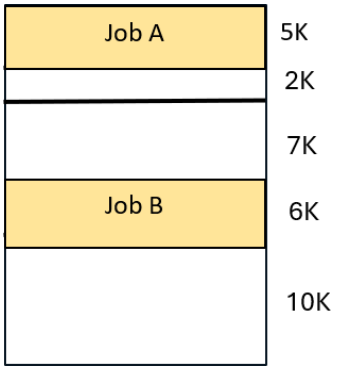


Figure 1

7. Figure 1 shows the main memory layout containing only Jobs A and B. The size of each segment is labelled on the right in Kilobytes. Assume that the dynamic partition approach is used. What will occur next if a new job with size 8K enter the memory?

A) The gap between Job A and Job B is closed.
B) The empty segments between Jobs A and B are merged.
C) Job A and Job B are rearranged to reduce the size of empty spaces.
D) Nothing will be changed on partitions.

8. To allocate more space to new processes, idle processes are removed from the main memory into the secondary memory, a process known as swapping. What considerations need to be made regarding swapping?

i. CPU clock speed.
ii. Memory capacity
iii. I/O status
iv. Cache size

A) i and ii
B) ii and iii
C) iii and iv
D) i and iv

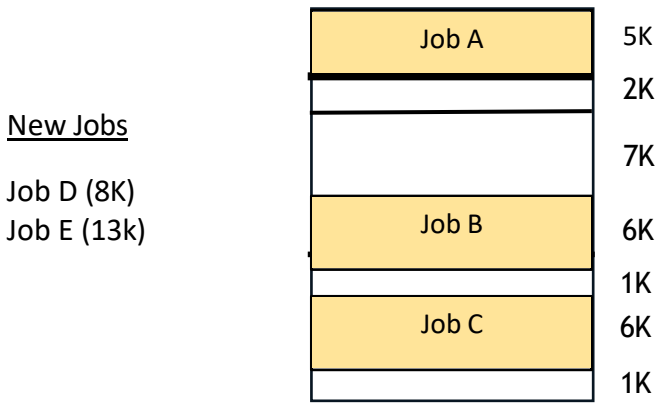


Figure 2

9. Figure 2 shows the fixed memory allocation layout containing Jobs A, B and C, with the size of each segment labelled on the right. New jobs D and E with the required segment size in the brackets cannot be allocated to the main memory because of _____.

A) External Fragmentation
B) Internal Fragmentation
C) Insufficient partition
D) Jobs A, B and C are still running.

10. _____ translates the logical address into a physical address.

A) MMU
B) Translator
C) Compiler
D) Pager

11. The page required to be replaced which algorithm chooses the page that has not been used for the longest period?

A) Optimal algorithm page replacement algorithm
B) Least Recently Used page replacement algorithm
C) Counting Based page replacement algorithm
D) First In First Out page replacement algorithm

12. Internal fragmentation occurs when a _____

A) memory area remains unused because it is too large to be allocated.
B) memory area remains unused because it is too small to be allocated.
C) more memory is allocated than requested by the process.
D) less memory is allocated than requested by the process.

13. In a file system utilizing the File Allocation Table (FAT) method for linked allocation, what does a directory entry typically contain?

A) Disk block addresses of file data.
B) File name, size, and attributes.
C) File creation timestamp.
D) File access permissions

14. In a file-system structure, which component maintains information about files such as their names, sizes, and locations on disk?

A) Disk allocator
B) File descriptor
C) File allocation table
D) Directory structure

15. Based on figure 3 below, what will happen the value for i, ii, iii and iv in physical memory?

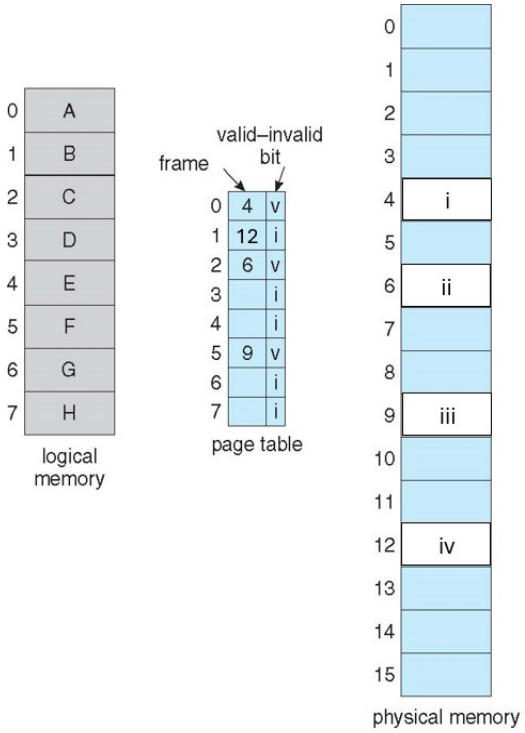


Figure 3

A) i = A, ii = B, iii = C, iv = D

B) i = D, ii = C, iii = B, iv = A

C) i = A, ii = C, iii = F, iv = H

D) i = A, ii = C, iii = F, iv = B

16. Which of the following disk scheduling algorithms can potentially lead to starvation?

A) First-Come First-Served (FCFS)

B) Shortest Seek Time First (SSTF)

C) SCAN

D) C-SCAN

17. Which of the following statements accurately describe RAID 6?

i. It uses two independent parity schemes.
ii. It can tolerate the failure of any two disks.
iii. It provides faster write performance compared to RAID 5.
iv. It requires at least four disks to implement.

A) i, ii, and iv
B) ii, iii, and iv
C) i, iii, and iv
D) i, ii, iii, and iv

18. Given a disk with 400 cylinders and the following sequence of requests [23, 89, 132, 42, 187, 299, 87], starting from cylinder 100, what is the next track location if the C-SCAN algorithm is used with the direction inner→outer.

A) 23
B) 89
C) 187
D) 132

19. In a file system utilizing modified contiguous allocation, what technique is employed to mitigate external fragmentation?

A) File compression
B) Extent allocation
C) Fragment reorganization
D) Disk defragmentation

20. In the context of free-space management in file systems, what does the counting method involve?

A) Calculating the total number of disk blocks on the storage device.
B) Utilizing linked lists to track free disk blocks.
C) Tracking the number of free disk blocks using a bitmap.
D) Maintaining a list of all free disk blocks.

PART B: STRUCTURED

Answer ALL questions in the answer booklet.

QUESTION 1 [16 MARKS]

The set $\{PP_{ii}, RR_{nn}\}$ denotes the relationship between a process and a resource where process PP_{ii} is requesting resource RR_{nn}.

The set $\{RR_{mm}, PP_{jj}\}$ denotes the relationship between a resource and a process where resource RR_{mm} is assigned to process PP_{jj}.

a. Given a set of vertices V = $\{PP_1, PP_2, PP_3, PP_4, RR_1, RR_2, RR_3, RR_4\}$ and set of edges E = { $\{PP_1, RR_1\}$, $\{PP_2, RR_3\}$, $\{PP_3, RR_1\}$, $\{PP_4, RR_2\}$, $\{RR_1, PP_2\}$, $\{RR_2, PP_3\}$, $\{RR_3, PP_4\}$, $\{RR_4, PP_1\}$ }. Sketch the resource allocation graph. **[5 marks]**

b. Is the system in a deadlock? Justify your answer. **[3 marks]**

c. If the system is in a deadlock, suggest a resource that a process should release. **[2 marks]**

d. Sketch the wait-for graph. **[2 marks]**

e. Given the table below, use a suitable algorithm to check if the system is in a deadlock. **[4 marks]**

	Allocation			**Request**			**Available**		
	A	B	C	A	B	C	A	B	C
P0	0	3	0	2	0	1	0	0	0
P1	1	1	1	1	0	3			
P2	2	0	3	0	0	0			
P3	4	1	0	1	2	5			

QUESTION 2 [16 MARKS]

Figure 4 shows the memory layout that contains empty segments represented by the shaded blocks. The numbers on the right of the memory layout are the addresses in Kilobytes. Answer questions (a) to (c).

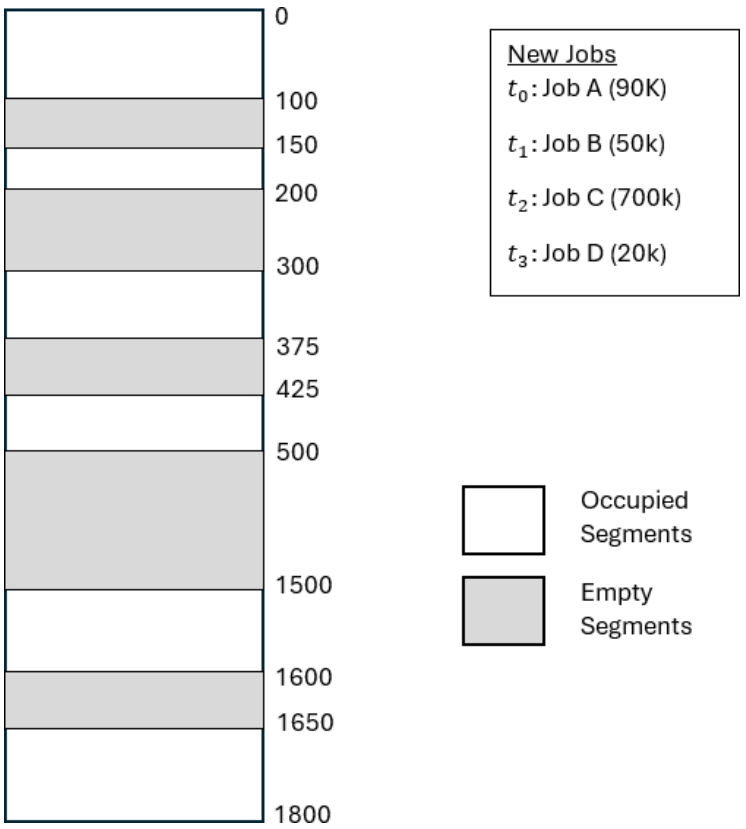


Figure 4

a. Four jobs A, B, C and D request memory allocation with the required size in the bracket. Draw the new memory layout using the First-Fit and Next-Fit memory allocation approaches once all jobs have been allocated to the memory. **[8 Marks]**

b. From the memory layout drawn in (a), state each process's base and limit registers based on the Next-Fit memory allocation approach. **[4 Marks]**

c. Discuss the differences (or similarities) between the first fit and the next fit. Suggest a better memory allocation approach. **[4 Marks]**

QUESTION 3 [16 MARKS]

Consider a system with 4-page frames and three-page replacement algorithms: FIFO (First-In-First-Out), OPT (Optimal), and LRU (Least Recently Used). Given a reference string as below:

1	2	3	4	1	2	5	6	1	2	0	3	4	6	0	2

Calculate how many page faults will occur:

a. If FIFO page replacement with 4-page frames is used. Show all your workings. [5]
b. If OPT page replacement with 4-page frames is used. Show all your workings. [5]
c. If LRU page replacement with 4-page frames is used. Show all your workings. [5]
d. From the above results, which has the best performance and discuss? [1]

QUESTION 4 [16 MARKS]

a. Consider the indexed allocation is used in a system to manage the following disk-space:

3, 16, 7, 19, 4, 5, 9, 12, 2, 10, 15, 8, 20, 1, 6, 11, 14, 23, 13

Assume each block can only store five block addresses with a linked scheme. Draw all the blocks for the disk-space. [5 *marks*]

b. Figure 5 shows a portion of bit map for free space management for a hypothetical Operating System whereby bit 0 and 1 indicates that the block is used and free, respectively. Suppose the shaded block indicates the 4th word for the bit map.

Block No.	0	1	2	3	…													
Bit Map :	0	0	0	0	…	0	0	0	0	0	0	0	1	0	1	1	0	…

Figure 5

Find the address of the first free block. Show your working. [6 *marks*]

c. Consider the following Figure 6 of a bit map for a disk of 32 blocks labeled from 0 to 31.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	0	0	1	1	1	0	1	0	0	0	1	1	1	1	0

16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
1	1	1	1	1	1	1	0	0	1	1	1	1	1	1	1

Figure 6

Produce the free-block of the disk using counting technique. [5 *Marks*]

QUESTION 5 [16 MARKS]

a. Given the disk scheduling scenario outlined below, calculate the total distance the disk arm moves using different algorithms and illustrate the movement using graphs:

- A disk drive with 300 cylinders numbered from 0 to 299.
- A queue of requests for disk arm movement at the following cylinders:
- [55, 147, 223, 89, 176, 90, 63, 105].
- The initial position of the disk arm is at cylinder 120.

For the following disk scheduling algorithms, determine the total distance the disk arm travels to satisfy all the requests and illustrate this movement with a graph. Make sure to label the x position for each point and provide a wide enough view to include all graph points. Provide the calculation for the total distance travelled for each algorithm.

i. Shortest Seek Time First (SSTF) **[7 marks]**

ii. C-SCAN (outward movement, inner→outer) **[7 marks]**

b. Based on your calculations in part (a), which algorithm is most efficient in terms of total head movement? Justify your answer. **[2 mark]**